
STATISTICAL MACHINE LEARNING

Homework Assignment 1

Due Date: Sunday, December 8, 2024 - By 11:59pm

EXERCISE 1: (57 POINTS)

Part 1: Dataset Exploration

1. Download the file `aims-sml-2024-2025-data.csv` and load it into RMarkdown.
2. Determine the size of the dataset n .
3. Create a scatterplot of y versus x .
4. Determine whether this is a classification or regression task, and justify your answer.

Part 2: Theoretical Framework

1. Suggest a function space $\mathcal{H}$ for this task and write it in mathematical notation.
2. Specify the loss function for this task and justify its use.
3. Derive and write down the theoretical risk $R(f)$ for a candidate function $f \in \mathcal{H}$.
4. Write down the expression for the Bayes learning machine $f^*(x)$ in this case.
5. Write down the empirical risk $\hat{R}(f)$ for a candidate function, including all components of the model space. *Hint: Decide on a complexity prior to training.*

Part 3: Estimation and Model Complexity

1. Derive the expression for the OLS estimator $\hat{f}(x)$ for this problem.
2. Comment extensively on the properties of $\hat{f}(x)$.
3. Use V -fold cross-validation (e.g., $V = 5, 10$) to determine the optimal complexity (degree p) for the polynomial regression model. Explain what “optimal complexity” means.
4. Plot the cross-validation error and empirical risk as functions of p . Comment on the plot.

Part 4: Model Comparison and Evaluation

1. Fit and plot the following models on the same plot with the data:

- The simplest estimator $\hat{f}(x)$ that depends on x .
- The optimal estimator determined by cross-validation.
- An overly complex model.

Use a legend to distinguish the models and comment on their behaviors.

2. Perform stochastic hold-out validation with $S = 100$ splits (70% training, 30% testing). Compute and plot boxplots of the test errors for:

- The simplest model.
- The optimal model.
- The overly complex model.

Part 5: Further Analysis

1. Perform an analysis of variance (ANOVA) on the test errors. What does it reveal?

2. Obtain and plot the 95% confidence and prediction bands for the dataset D_n .

3. Write the mathematical expression for:

- The confidence band for a single observation (X_i, Y_i) .
- The prediction band for a single observation (X_i, Y_i) .

4. Comment extensively on what the confidence and prediction bands reveal about the model.

EXERCISE 2: (32 POINTS)

Consider the `spam` dataset from `library(kernlab)`. You are supposed to provide a thorough comparison of four learning machines namely LDA, QDA, Naive Bayes and FLD, and your comparison will be solely based on the test error.

1. Plot the distribution of the response for this dataset and comment.
2. Comment on the shape of this dataset in terms of the sample size and the dimensionality of the input space
3. Comment succinctly from the statistical perspective on the type of data in the input space
4. Using the whole data for training and the whole data for test, building the above four learning machines, then plot the comparative ROC curves on the same grid

5. Comment succinctly on what the ROC curves reveal for this data and argue in light of the theory whether or not that was to be expected.
6. Using `set.seed(19671210)` along with a 2/3 training 1/3 test in the context stratified stochastic holdout split of the data, compute $S = 50$ replications of the test error for all the above learning machines.
7. Plot the comparative boxplots (be sure to properly label the plots)
8. Comment on the distribution of the test error in light of (implicit) model complexity.

EXERCISE 3: (11 POINTS)

This video component of the homework assignment cover 11% of the score, and is therefore of vital importance. You must create a video summarizing your understanding of the spirit and substance of the each of the above exercises.

1. The video should be exactly 5 minutes long.
2. The video should feature you speaking directly with your face showing either in full or at the corner of your document.
3. The video should cover the gist of your understanding of the key spirit of each exercise.
4. Your completed pdf file should be displayed as you are commenting on it.
5. Your completed video should be uploaded to your private YouTube channel and a permission given to `tutors2024-25@aims.ac.rw` or made public if you feel comfortable doing so. At any rather your video should be on Youtube and accessible for marking/grading!
6. You should strive to demonstrate/showcase the understanding you gathered from doing the homework assignment.
7. A link to your Youtube video should be provided in the comment portion of myCourses where you submit your final report.

All your answers must be written on a separate sheet, properly typeset and submitted in the form of a report, in pdf format. No MS Word report will be accepted. No handwritten report will be accepted either, and I mean none. The use of RMarkdown is strongly encouraged! You are strongly encourage to use artificial intelligence tools as much as you want, but mindlessly copied outputs from AI chatbots will result in pure zero. A video component of this assignment will count for 11% of your score. Make sure your report is in pdf format and your video is mp4 with a clickage link to Youtube without your pdf or otherwise. All must be submitted by the deadline.